LTREB_SHIMADZUGC_CH4_TEMPLATE

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Contents

1 Read in data files **3**

2 Assess standard curves **4**

3 Aanlayze the Check Standards **6**

4 Analyze the Blanks **7**

5 Dilution correct samples **8**

6 Calculate gas in water **10**

7 Merge samples with Metadata & check all samples are present **10**

8 Visualize Data **10**

##Run Information

```r
cat("Run Information: Brenden D. Blakley ") #lets you know what section you're in
```

```
## Run Information: Brenden D. Blakley
```

```r
#set the run date & user name
  run_date <- "06/12/2025"
  sample_year <- "2025"
  sample_month <- "May"
  user <- "Brenden Blakley"

#identify the files you want to read in
  #read in as a list to accommodate multiple runs in a month

GHG_file <- "Raw Data/20250612_2M_MAY_GHG.csv"


# Define the file path for QAQC log file - NO Need to change just check year
log_path <- "Processed Data/LTREB_2M_ShimadzuGC_QAQC_2025.csv"

# Define final path for data be sure to change year and month and run number
final_path <- "Processed Data/GCReW_CO2xN_Porewater_CH4_202505_1.csv"
#Gas in Water Calculations
  # "TempC" column value in Kelvin (equilibration temperature) and volume of gas (in liters) in syringe
TempC = 22.7
volume = 0.012

#record any notes about the run or anything other info here:
  run_notes <- ""

#Set up file path for metadata
  #downloaded metadata csv - downloaded from Google drive as csv for this year
  Raw_Metadata = "Raw Data/Sample_IDs_LTREB_SMARTX_2024.csv"
  #Collection_Dates = "Raw Data/Sample_Collection_Dates_2024.csv"

  cat(run_notes)
```

##Set Up

##Set Up Constants

```r
##Low & High Standard Curve Cutoffs
low_std_crv_cutoff <- 1000
low_std_crv_area_cutoff <- 600000
r2_cutoff <- 0.990

##Gas In Water Calculation



##Detection Limits
adl_cutoff <- 20000
```

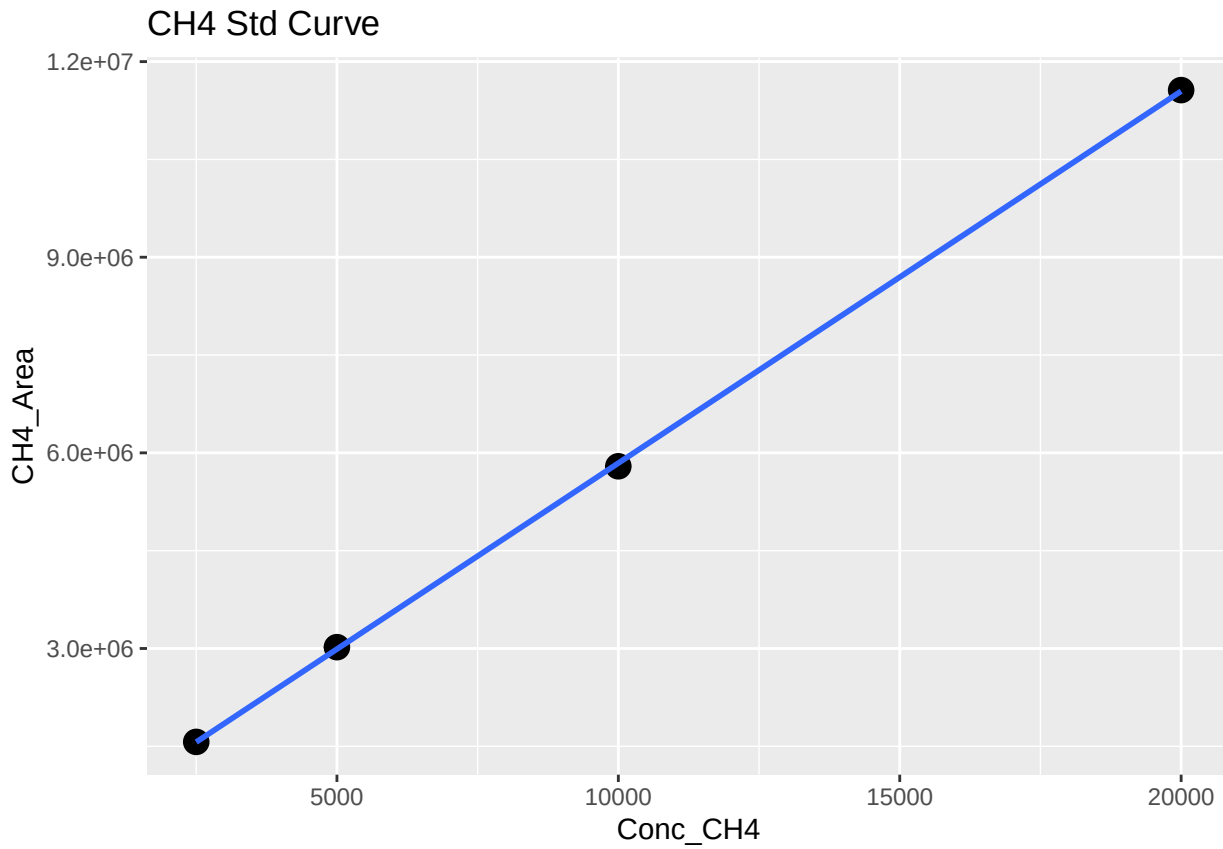
```r
bdl_cutoff <- 50

##Check Standard Concentration
level_ch4 <- 1000      #ppm
chks_flag = 60
```

##Read in metadata and create similar sample IDs for matching to samples

1 Read in data files

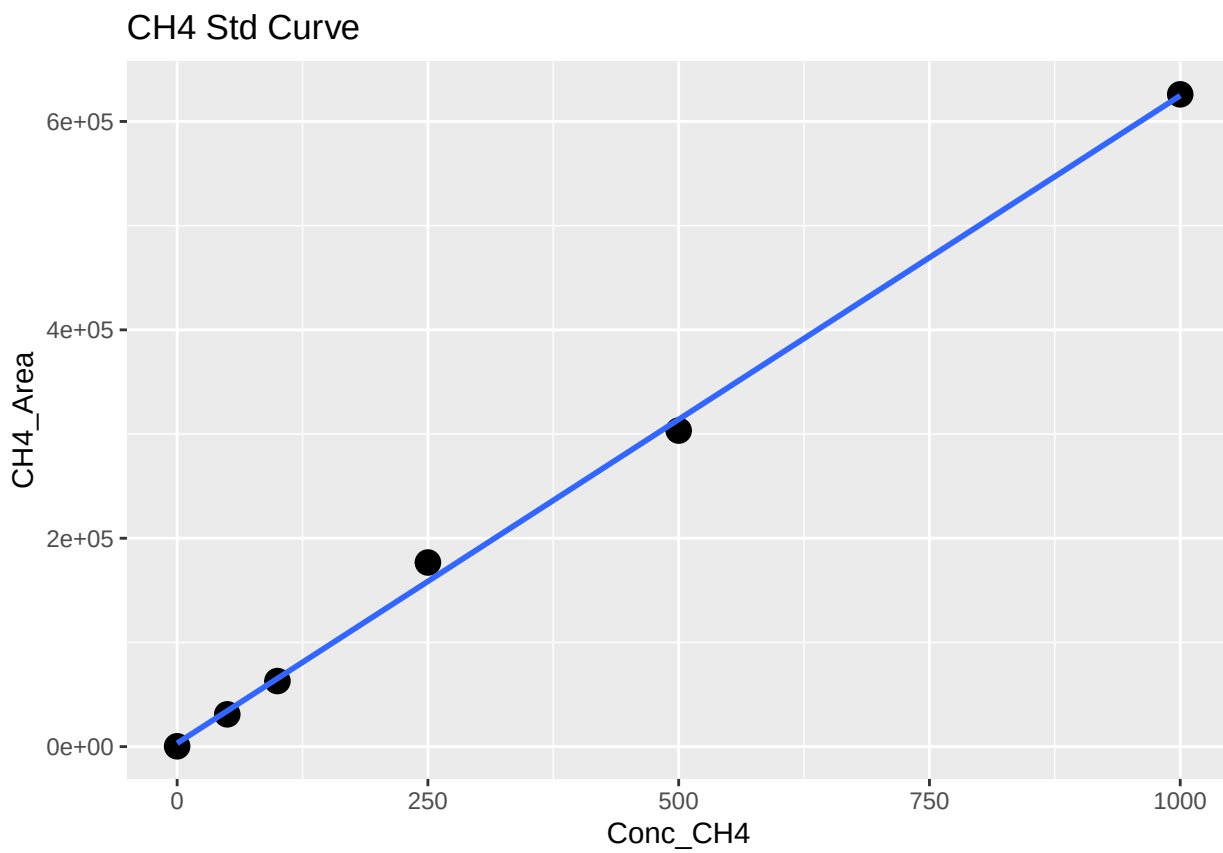
2 Assess standard curves

```
## `geom_smooth()` using formula = 'y ~ x'
```



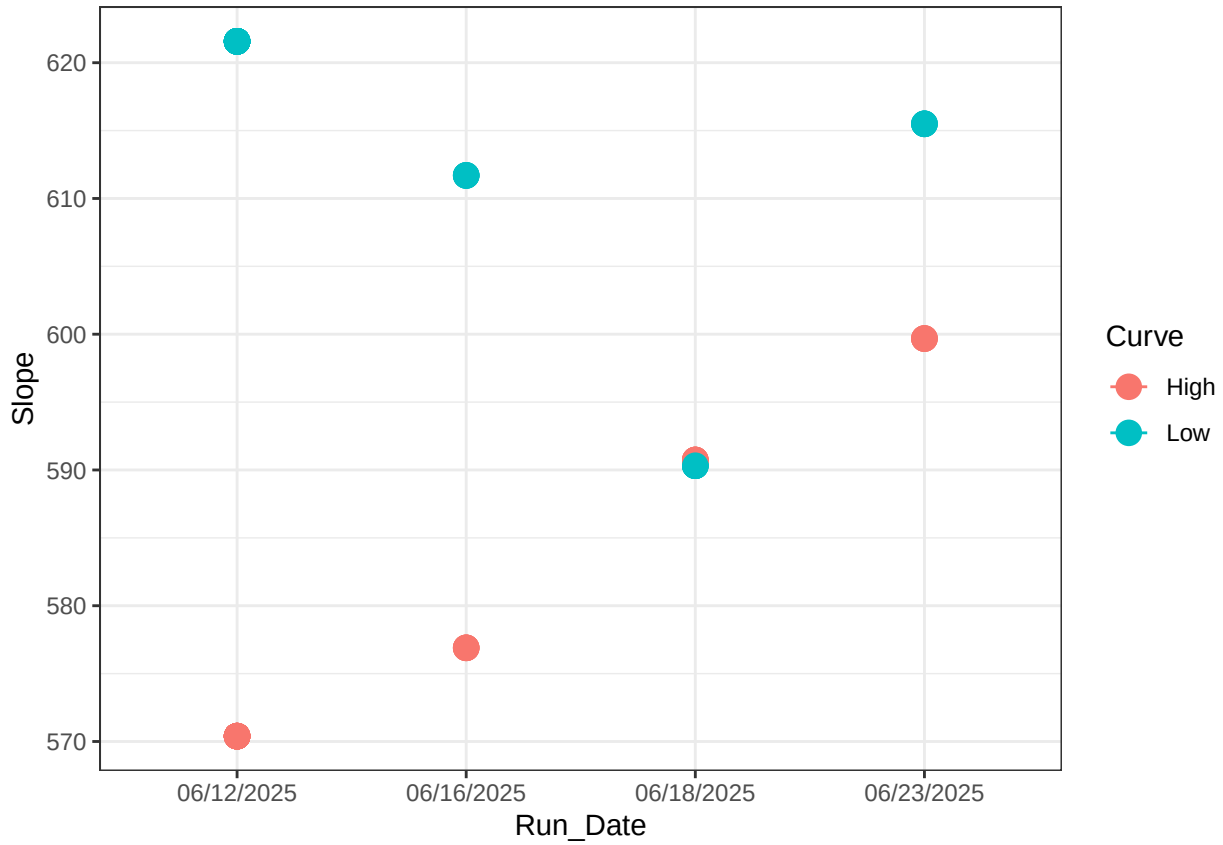

```
##
## Call:
## lm(formula = high_stds_ch4$CH4_Area ~ high_stds_ch4$Conc_CH4)
##
## Residuals:
##      1      2      3      4
##   2655  29301 -48599  16642
##
## Coefficients:
##                         Estimate Std. Error t value Pr(>|t|)
## (Intercept)            1.387e+05  3.599e+04   3.855   0.0612 .
## high_stds_ch4$Conc_CH4 5.704e+02  3.123e+00 182.658    3e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 41860 on 2 degrees of freedom
## Multiple R-squared:  0.9999, Adjusted R-squared:  0.9999
## F-statistic: 3.336e+04 on 1 and 2 DF,  p-value: 2.997e-05
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```
##
## Call:
## lm(formula = low_stds_ch4$CH4_Area ~ low_stds_ch4$Conc_CH4)
##
## Residuals:
##      1      2      3      4      5      6
##  -2894  -3297  -2444  18089 -10683   1228
##
## Coefficients:
##                        Estimate Std. Error t value Pr(>|t|)
## (Intercept)             3224.25    5977.12   0.539    0.618
## low_stds_ch4$Conc_CH4    621.57      12.72  48.869 1.05e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 10820 on 4 degrees of freedom
## Multiple R-squared:  0.9983, Adjusted R-squared:  0.9979
## F-statistic:  2388 on 1 and 4 DF,  p-value: 1.049e-06
```

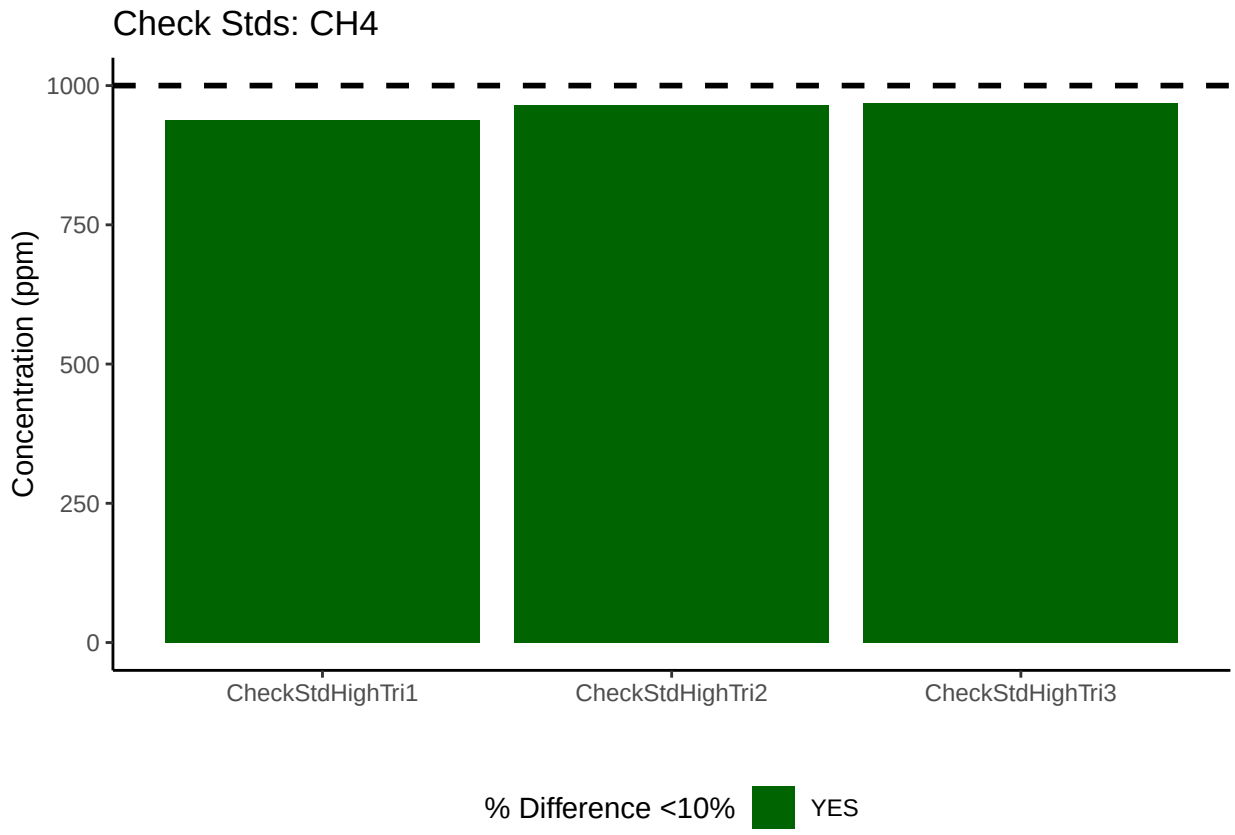
```
##   X Curve        R2    Slope    Intercept   Run_Date
## 1 1  High 0.9986295 576.8990  173986.2174 06/16/2025
## 2 2   Low 0.9990260 611.6961   -3791.2684 06/16/2025
## 3 3  High 0.9999101 570.3925  138747.3478 06/12/2025
## 4 4   Low 0.9979098 621.5734    3224.2535 06/12/2025
## 5 5  High 0.9989882 590.7374 -109278.7391 06/18/2025
## 6 6   Low 0.9990995 590.3138     954.7682 06/18/2025
```



2.1 Now calculate the CH4 & CO2 concentrations in ppm

3 Aanlayze the Check Standards

```
## Warning: Using ‘size‘ aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use ‘linewidth‘ instead.
## This warning is displayed once every 8 hours.
## Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
## generated.
```



4 Analyze the Blanks

```
## Assess Blanks
```

```
## [1] ">60% of CH4 Blanks are below the lower 25% quartile of samples - PROCEED"
```

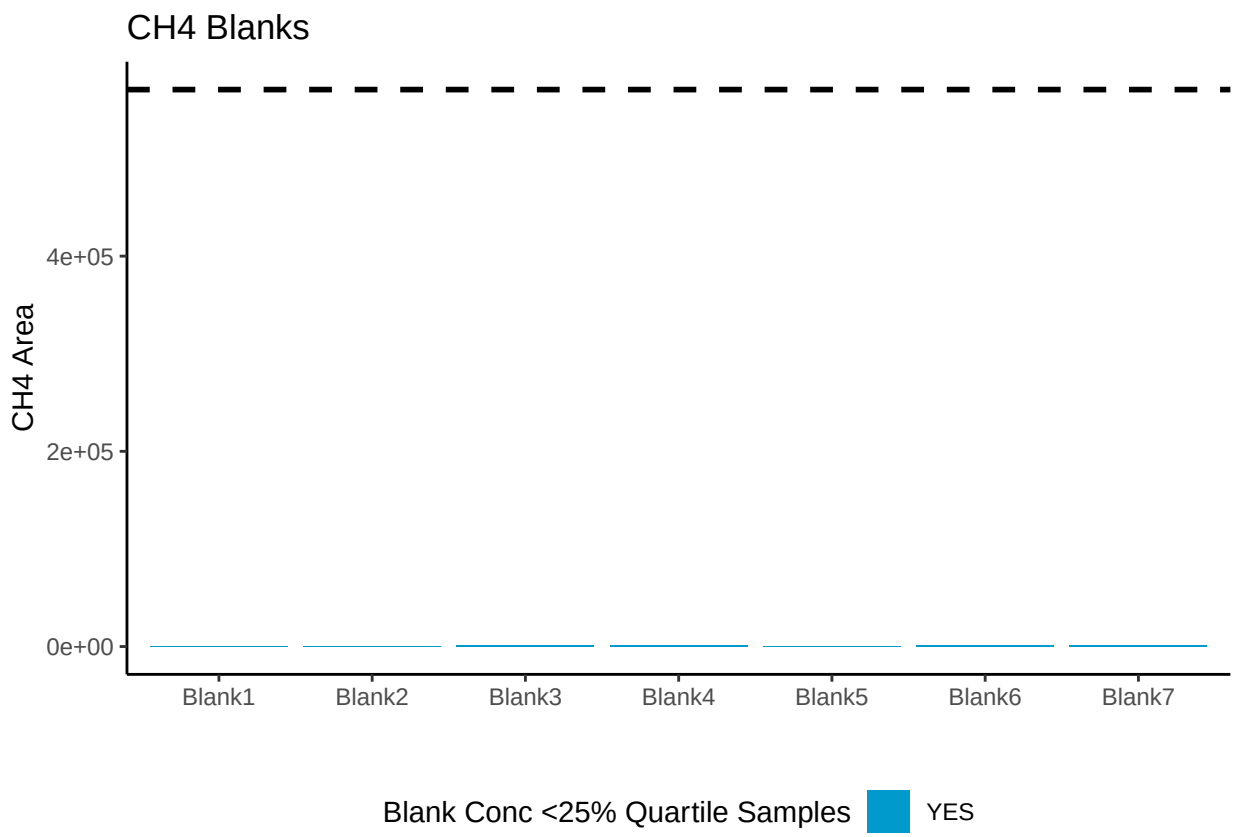
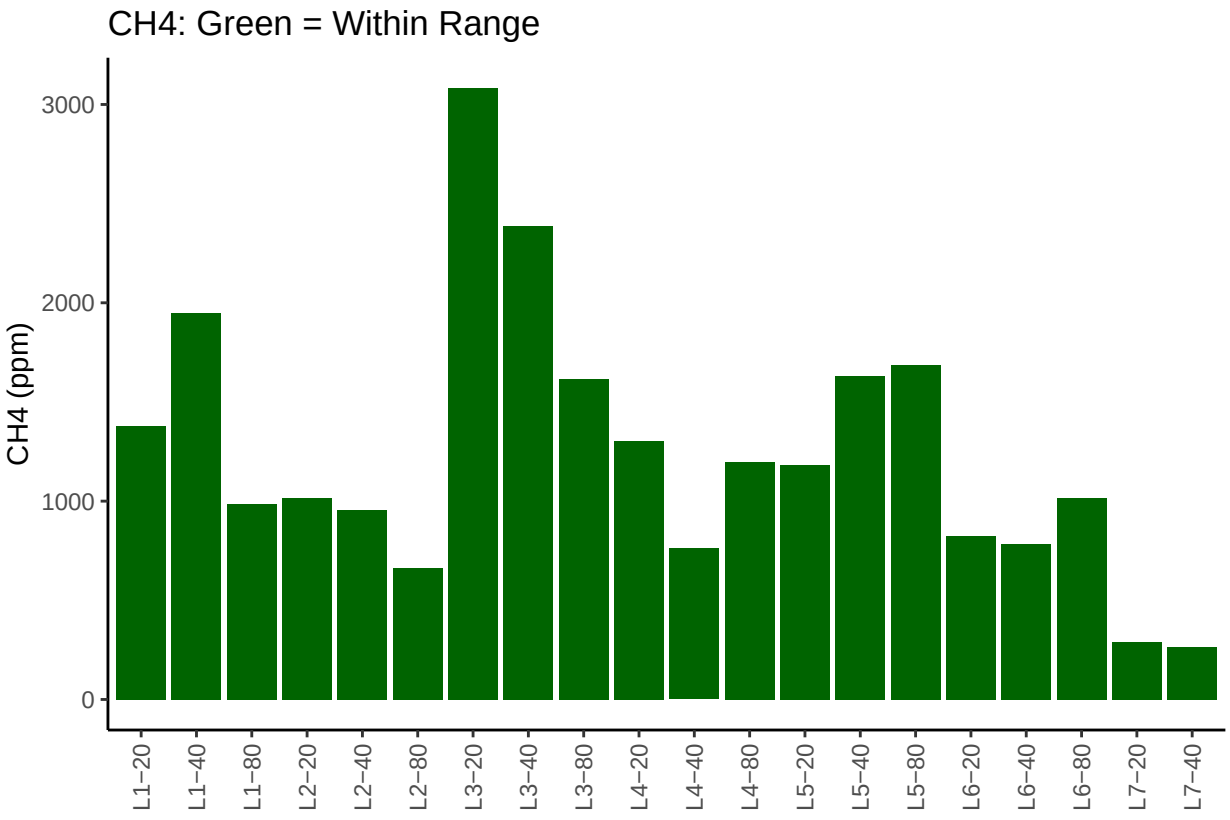


Table 1: Mean Concentration of Blanks

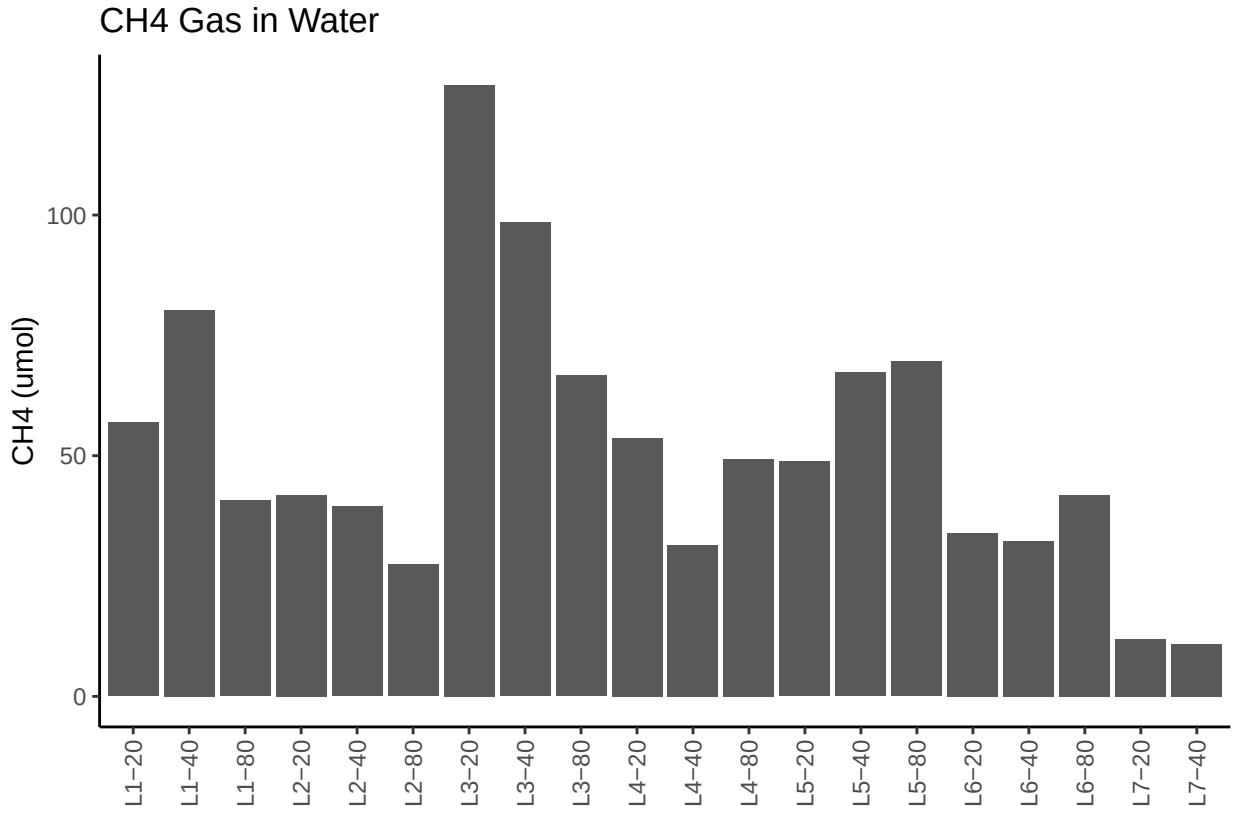
Test	Blank_Mean_Area
CH4	1294.143

5 Dilution correct samples

```
#multiply the concentration by the dilution factor
Samples$CH4_Conc_ppm_dilcorr <- (Samples$CH4_Conc_ppm * 12 + Samples$Sample_Volume )/(Samples$Sample_Vo

#check results
#head(Samples)

#quick first look at the samples
ch4_samples <-  ggplot(data = Samples, aes(x = Sample_ID, y = CH4_Conc_ppm, fill=CH4_Flag)) +
       geom_bar(stat = 'identity') +
        scale_fill_manual(values=c("darkgreen","red"))+
        theme_classic() + labs(x= " ", y="CH4 (ppm)", title="CH4: Green = Within Range") +
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1)) +
        theme(legend.position="none")

ch4_samples
```


CH4: Green = Within Range
CH4 (ppm)
3000
2000
1000
0
L1−20
L1−40
L1−80
L2−20
L2−40
L2−80
L3−20
L3−40
L3−80
L4−20
L4−40
L4−80
L5−20
L5−40
L5−80
L6−20
L6−40
L6−80
L7−20
L7−40

6 Calculate gas in water



7 Merge samples with Metadata & check all samples are present

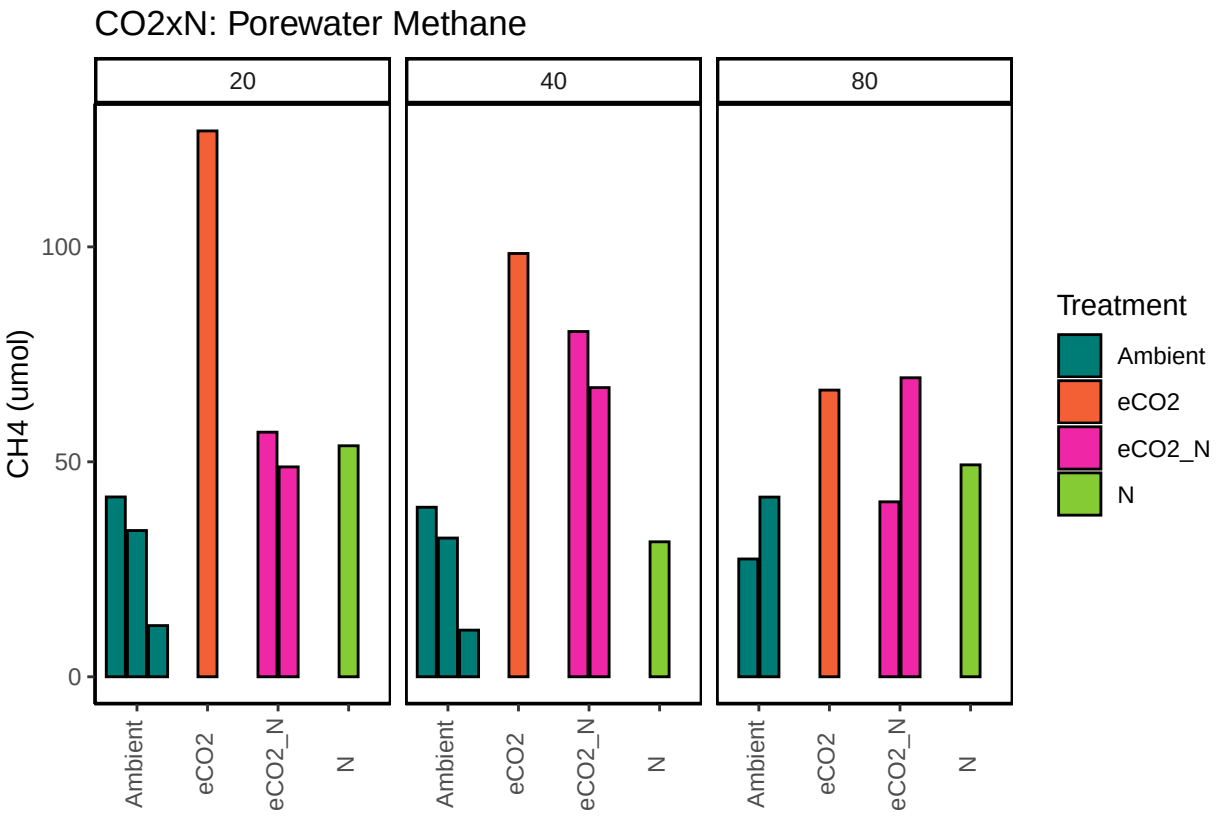
```
## Merge Samples with Metadata
```

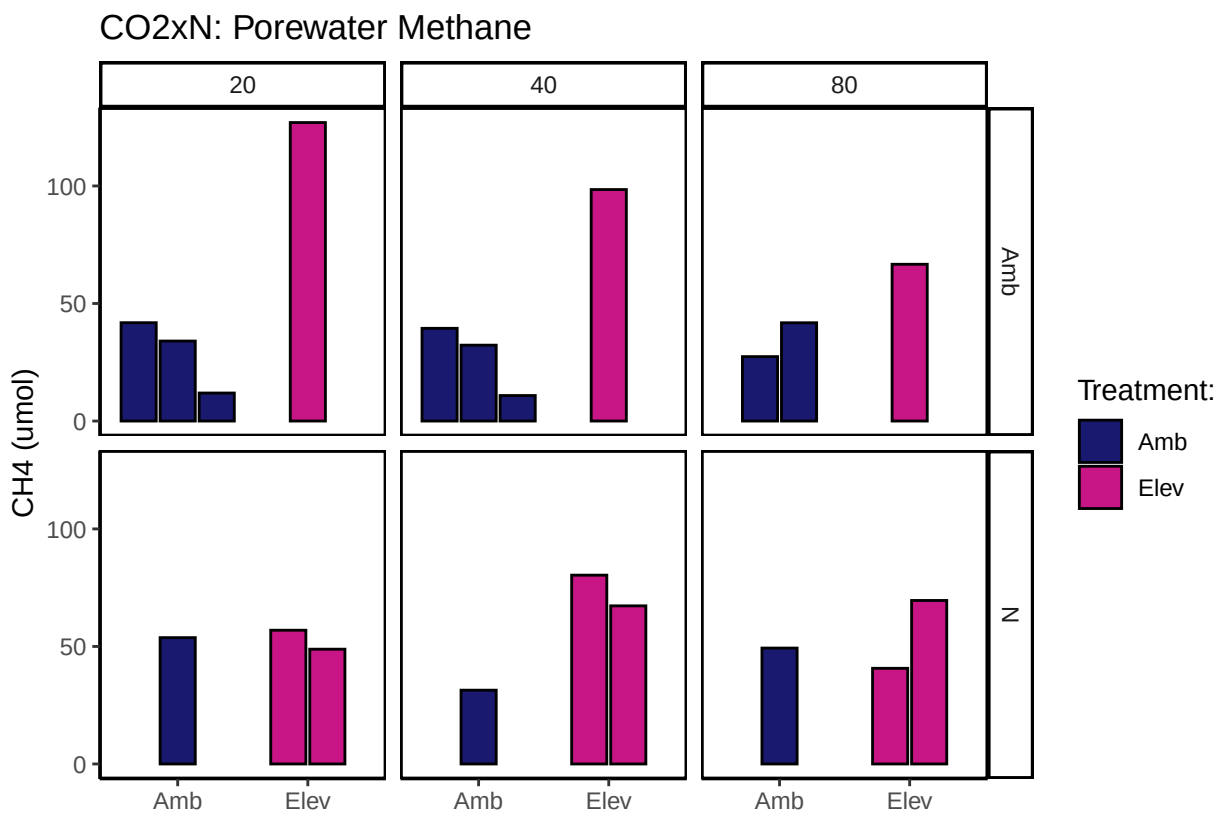
```
## Some sample IDs are missing from data.
```

```
##  [1] "L7-80"  "L8-20"  "L8-40"  "L8-80"  "L9-20"  "L9-40"  "L9-80"  "L10-20"
##  [9] "L10-40" "L10-80" "L11-20" "L11-40" "L11-80" "L12-20" "L12-40" "L12-80"
## [17] "L13-20" "L13-40" "L13-80" "L14-20" "L14-40" "L14-80" "L15-20" "L15-40"
## [25] "L15-80" "L16-20" "L16-40" "L16-80" "L17-20" "L17-40" "L17-80" "L18-20"
## [33] "L18-40" "L18-80" "L19-20" "L19-40" "L19-80" "L20-20" "L20-40" "L20-80"
```

8 Visualize Data

```
## Visualize Data
```


CO2xN: Porewater Methane
20
40
80
CH4 (umol)
100
50
0
Ambient
eCO2
eCO2_N
N
Treatment
Ambient
eCO2
eCO2_N
N



8.1 Export Processed Data

```
#check results
#head(Samples)

#pull out what we need
Samples1 <- df_all_clean[ ,c(7, 9:14, 16:23 )]
#head(Samples1)

#write.csv(Samples1, final_path)
```

#end